

ASSIGNMENT-10.5

Lab 10 – Code Review and Quality: Using AI to Improve Code

Quality and Readability

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Batch: 19

Task 1: Improving Variable & Function Names Original Code

```
def f(a, b):    return  
  
a + b print(f(10,  
  
20))
```

CODE:

```
def add_numbers(first_number, second_number):  
  
    """  
  
    Returns the sum of two numbers.  
  
    """  
  
    return first_number + second_number  
  
  
result = add_numbers(10, 20) print(result)
```

OUTPUT

```
PS C:\Users\trina\OneDrive\Desktop\python am> & "C:/Program Files/Python312/python.exe" "c:/Users/tri  
na/OneDrive/Documents/AI/10.5 task 1.py"  
30
```

Task Description #2 – Missing Error Handling

Task: Use AI to add proper error handling.

Sample Input Code: def

```
divide(a, b): return a / b
```

```
print(divide(10, 0))
```

Expected Output:

- Code with exception handling and clear error messages

CODE :

```
def divide_numbers(dividend, divisor):  
  
    """  
  
    Divides two numbers safely.  
  
    """  
  
    try:  
  
        return dividend / divisor    except  
  
ZeroDivisionError:  
  
        return "Error: Division by zero is not allowed."  
  
    except TypeError:    return "Error: Please enter valid  
  
numbers."  
  
  
result = divide_numbers(10, 0) print(result)
```

OUTPUT:

```
PS C:\Users\trina\OneDrive\Desktop\python am> & "C:/Program Files/Python312/python.exe" "c:/Users/tri  
na/OneDrive/Documents/AI/10.5 task 1.py"  
Error: Division by zero is not allowed.
```

Task Description #3: Student Marks Processing System

The following program calculates total, average, and grade of a student, but it has poor readability, style issues, and no error handling.

```
marks=[78,85,90,66,88] t=0  
  
for i in marks:  
  
    t=t+i  
  
a=t/len(marks) if  
  
a>=90: print("A")
```

```
elif a>=75:
```

```
print("B") elif
```

```
a>=60: print("C")
```

```
else:
```

```
print("F")
```

Task:

- Use AI to refactor the code to follow PEP 8 standards.
- Add meaningful variable names, functions, and comments.
- Add basic input validation and documentation.

CODE :

```
def calculate_average(marks_list):
```

```
    """
```

```
    Calculates the average of student marks.
```

```
    Args:
```

```
        marks_list (list): List of student marks
```

```
    Returns:
```

```
        float: Average marks
```

```
    """
    if not marks_list:
        raise ValueError("Marks list
cannot be empty.")
```

```
    return sum(marks_list) / len(marks_list)
```

```
def determine_grade(average_marks):
```

```
    """
```

```
    Determines grade based on average marks.
```

```
    """
```

```

if average_marks >= 90:

    return "A"    elif

average_marks >= 75:

return "B"    elif

average_marks >= 60:

    return "C"    else:

    return "F"


def main():

    """

    Main function to process student marks.

    """

    marks = [78, 85, 90, 66, 88]

    try:

        average = calculate_average(marks)    grade =

determine_grade(average)

        print(f"Average Marks: {average:.2f}")

print(f"Grade: {grade}")

    except ValueError as error:    print(f"Error:

{error}")

if __name__ == "__main__":

    main()

```

OUTPUT:

```

PS C:\Users\trina\OneDrive\Desktop\python am> & "C:/Program Files/Python312/python.exe" "c:/Users/tri
na/OneDrive/Documents/AI/10.5 task 1.py"
Average Marks: 81.40
Grade: B

```

Task Description #4: Use AI to add docstrings and inline comments to the

following function. def factorial(n):

result = 1 for i in

range(1,n+1):

result *= i

return result **CODE :**

```
def factorial(number):
```

```
    """
```

```
    Calculates the factorial of a given number.
```

```
    """
```

```
    if number < 0:        raise ValueError("Negative numbers not
allowed.")
```

```
    result = 1
```

```
    for i in range(1, number + 1):
```

```
        result *= i
```

```
    return result
```

```
if __name__ == "__main__":
```

```
    number = int(input("Enter a number: "))    print("Factorial:",
factorial(number))
```

OUTPUT:

```
PS C:\Users\trina\OneDrive\Desktop\python am> & "C:/Program Files/Python312/python.exe" "c:/Users/tri
na/OneDrive/Documents/AI/10.5 task 1.py"
Enter a number: 10
Factorial: 3628800
```

Task Description #5: Password Validation System (Enhanced)

The following

Python program validates a password using only a minimum length check, which is insufficient for real-world security requirements.

```
pwd = input("Enter password: ") if
len(pwd) >= 8: print("Strong") else:

print("Weak")
```

CODE : import re

```
def is_strong_password(password):
```

```
    """
```

Checks whether the given password meets security standards.

Rules:

- Minimum 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one digit
- At least one special character

Args: password (str): User-entered

password

Returns: bool: True if strong, False

otherwise

```
    """
```

```

if len(password) < 8:

return False


if not re.search(r"[A-Z]", password):

    return False


if not re.search(r"[a-z]", password):

    return False


if not re.search(r"[0-9]", password):    return

False


if not re.search(r"[!@#$%^&*()_+{}[\];<>,.?~\|-]", password):    return

False


return True


def main():

    """

    Main function for password validation.

    """

    user_password = input("Enter password: ")

    if is_strong_password(user_password):

        print("Password is Strong ")    else:

        print("Password is Weak ")

    print("It must contain:")    print("- At

least 8 characters")    print("- One

```

```
uppercase letter")    print("- One
lowercase letter")    print("- One digit")

print("- One special character") if

__name__ == "__main__":

    main()
```

OUTPUT

```
PS C:\Users\trina\OneDrive\Desktop\python am> & "C:/Program Files/Python312/python.exe" "c:/Users/tri
na/OneDrive/Documents/AI/10.5 assignment(1264).py"
Enter password: haha
Password is Weak ❌
It must contain:
- At least 8 characters
- One uppercase letter
- One lowercase letter
- One digit
- One special character
PS C:\Users\trina\OneDrive\Desktop\python am> & "C:/Program Files/Python312/python.exe" "c:/Users/tri
na/OneDrive/Documents/AI/10.5 assignment(1264).py"
Enter password: Trinayani@123
Password is Strong ✅
```